

Fractional powers



1 Express the following in exponential form:

a $\sqrt{x}$

b $\sqrt[6]{x}$

c $\sqrt[5]{x^7}$

d $\sqrt[5]{5}$

e $\sqrt[7]{71}$

f $(\sqrt[3]{5})^5$

g $\sqrt[3]{x^{6a}}$

2 Express $\frac{1}{\sqrt{x}}$ with a negative fractional index.

3 Express the following in radical form:

a $3^{\frac{1}{2}}$

b $2^{\frac{4}{7}}$

4 Without using a calculator, evaluate the following:

a $121^{\frac{1}{2}}$

b $1^{\frac{1}{10}}$

c $4^{\frac{3}{2}}$

d $(-64)^{\frac{1}{3}}$

e $64^{-\frac{1}{6}}$

f $(64^{\frac{1}{9}})^{\frac{9}{2}}$

g $512^{\frac{1}{3}}$

h $625^{\frac{1}{4}}$

i $1024^{\frac{1}{5}}$

j $128^{\frac{1}{7}}$

k $36^{\frac{3}{2}}$

l $10\,000^{\frac{3}{4}}$

m $32^{\frac{2}{5}}$

n $64^{\frac{5}{6}}$

o $0^{\frac{8}{5}}$

p $(-64)^{\frac{4}{3}}$

q $32^{-\frac{2}{5}}$

r $(-243)^{-\frac{2}{5}}$

s $(4a^4)^{\frac{1}{2}}$

t $8^{\frac{5}{3}}$

5 Simplify the following:

a $\left(\frac{3^3}{512}\right)^{\frac{1}{3}}$

b $\left(\frac{25}{36}\right)^{\frac{1}{2}}$

c $\left(\frac{169}{16}\right)^{\frac{1}{2}}$

d $\left(\frac{512}{729}\right)^{\frac{1}{3}}$

e $\left(\frac{625}{81}\right)^{\frac{1}{4}}$

f $\left(\frac{4}{9}\right)^{\frac{5}{2}}$

g $\left(\frac{64}{27}\right)^{-\frac{1}{3}}$

h $\frac{1}{243^{-\frac{3}{5}}}$

i $\frac{2^{\frac{14}{3}}}{2^{\frac{2}{3}}}$

j $\frac{2^{\frac{4}{9}}}{2^{\frac{7}{9}}}$

k $\left(\frac{x^{15}}{1024}\right)^{\frac{2}{5}}$

6 Simplify the following expressions and then evaluate:

a $36^{\frac{3}{2}} - 32^{\frac{7}{5}}$

b $8^{-\frac{4}{7}} \times 8^{-\frac{2}{7}}$

c $100^{-\frac{3}{4}} \times 100^{\frac{1}{4}}$

d $7^{\frac{1}{7}} \times 49^{\frac{4}{14}}$

7 Find the missing value in the following equation:

$$625^{\frac{3}{5}} = 125$$



8 Simplify the following expressions, giving your answers in exponential form:

a $(625u^{16}v^{12})^{\frac{1}{2}}$

b $y^3 \times \sqrt[3]{y}$

c $y^{\frac{6}{5}} \times y^2$

d $2y^{\frac{3}{5}} \times 2y^{\frac{2}{5}}$

e $4b^{\frac{3}{2}} \div 2b^{\frac{2}{3}}$

9 Simplify $\sqrt{2} \times \sqrt[3]{2}$ and give your answer in radical form.

10 To evaluate $81^{\frac{3}{2}}$ would it be more efficient to use the property $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$, or the property $a^{\frac{m}{n}} = \sqrt[n]{a^m}$. Explain your answer.

11 Simplify the following expressions, giving your answers in radical form:

a $\sqrt[8]{x^2}$

b $\sqrt[5]{\sqrt{x}}$

12 Simplify $\left(\sqrt[4]{x^5y^3}\right)^{24}$. Assume x and y represent positive numbers.

13 Consider the expression $p^{\frac{q}{r}}$. Complete the following statements:

- a One way of understanding this expression is to say that we are raising p to the power of $\frac{q}{r}$, then taking the r th root of the result.
- b Another way of understanding this expression is to say that we are taking the r th root of p^q , and then raising the result to the power of $\frac{1}{r}$.

Applications

14 The approximate radius, r , of a circle with an area, A , is $r = 0.32A^{\frac{1}{2}}$. Find the approximate radius of a circle with an area of 64 in^2 .

15 The surface area, A , of a cube with a volume, V , is $A = 6V^{\frac{2}{3}}$. Find the surface area of a cube with a volume of 1000 in^3 .

16 The volume, V , of a cube with a surface area, A , is $V = \left(\frac{A}{6}\right)^{\frac{3}{2}}$. Find the volume of a cube with a surface area of 54 in^2 .

17 A company's net profit (in dollars) can be modeled by the equation $P = 160n^{\frac{3}{2}} - 1100$, where n is the number of units sold. What will its net profit be if it manages to sell 16 units?